

Ants on a Stick: Collision Invariance and Exit-Time Bounds

Arian Nazarian

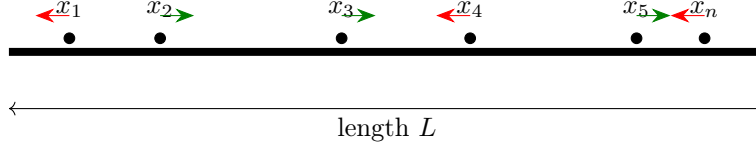


Figure 1: A finite system of identical agents on an interval. Each agent moves at speed v either left or right, reverses direction at collisions, and exits at an endpoint.

Abstract

How long can it take for a group of ants to fall off a stick if every ant moves at the same speed and reverses direction whenever it collides with another ant? The familiar puzzle has a deceptively simple answer: if the ants are indistinguishable, a collision can be treated as two ants passing through one another and exchanging labels. Thus the last exit time depends only on the farthest directed distance to an endpoint.

This paper develops that observation into a deterministic, algorithmic, and distributional analysis of an interacting-agent system on an interval. We formalize the collision-invariance principle, characterize the full multiset of exit times, prove maximum and minimum completion-time bounds, give optimal algorithms for completion time and ordered exit times, and count the number of possible physical collisions. Under random initial positions and directions, the exit process reduces to standard order statistics, yielding exact formulas for the completion-time distribution, expectation, variance, quantiles, and the number of agents remaining at a given time. A final section interprets the model as a stylized time-to-clear problem: not a literal financial-market model, but a compact example of how an apparently interaction-heavy system can collapse into a tractable bottleneck variable.

1 The Ants-on-a-Stick Problem

The classic ants-on-a-stick puzzle asks for the maximum time required for a group of ants to fall from a stick when all ants move at the same speed and reverse direction at collisions. A standard version asks about ten ants on a two-foot stick moving at one foot per minute. At first glance, the problem appears to require tracking a sequence of pairwise collisions. The key observation is that if the ants are identical, then a collision between two ants is equivalent to the ants passing through each other while exchanging labels. This is the familiar solution to the recreational puzzle, but it also suggests a more general modeling principle: some one-dimensional interacting systems have unlabeled observables that are invariant under collision dynamics.^{5,4}

The goal of this paper is not to rediscover the classic trick. Instead, the goal is to build a complete exit-time framework around it. The analysis proceeds in four layers. First, we formulate a deterministic model and prove the collision-invariance lemma. Second, we characterize not only the final completion time but the entire unordered exit-time process. Third, we show that the completion time can be computed by an optimal linear-time algorithm even though the physical collision count may be quadratic. Finally, we impose random initial conditions and derive exact distributional, asymptotic, and risk-style quantities.

For the motivating classroom version, the stick length is

$$L = 2 \text{ feet}$$

and the common speed is

$$v = 1 \text{ foot/minute.}$$

The maximum completion time is therefore two minutes. The sections below explain why this answer is independent of the number of ants and why the same model yields richer stochastic information when initial positions and directions are randomized.

2 Deterministic Model

Consider the interval

$$[0, L]$$

with length $L > 0$. There are n point agents, called ants, initially located at positions

$$x_1, x_2, \dots, x_n \in [0, L].$$

Each ant moves at common speed $v > 0$ and has initial direction

$$d_i \in \{-1, +1\},$$

where $d_i = -1$ denotes motion to the left and $d_i = +1$ denotes motion to the right. Ants exit the system when they reach either endpoint. When two ants collide, they instantaneously reverse directions and continue moving at speed v .

The main assumptions are:

- ants are modeled as point particles;
- all ants have the same speed v ;
- collisions are instantaneous and elastic in the sense of direction reversal;
- labels are not relevant to the system completion time;
- simultaneous collisions are excluded in the base model, or handled as limits of configurations with distinct collision times.

The equal-speed assumption is essential. If agents have different speeds and retain their speeds after collision, the simple label-swap equivalence can fail. The present paper studies the equal-speed case because that is the setting in which the collision process admits an exact reduction.

3 Collision Invariance

The mathematical center of the problem is the observation that for identical ants, a collision can be removed by relabeling.

Lemma 1 (Collision invariance). *Consider two identical ants moving at equal speed on a line. The evolution in which the ants collide and reverse directions produces the same unordered set of future positions as the evolution in which the ants pass through each other and exchange labels.*

Proof. Immediately before a collision, one ant approaches from the left and moves right, while the other approaches from the right and moves left. In the physical collision model, both ants reverse direction at the meeting point. In the pass-through model, both ants continue moving through the meeting point. Since the ants have equal speed and are identical, the set of occupied positions after the meeting is the same in both descriptions. The only difference is the label assigned to each trajectory. If labels are ignored, collision-and-reversal and pass-through motion are equivalent. $\square$

This lemma is local, but applying it at every collision removes the entire interacting structure. The physical system can therefore be analyzed through a noninteracting pass-through system, provided the quantity of interest depends only on the set of occupied positions or exit times and not on the identity of a particular named ant.

4 Exit Times and Extremal Bounds

For each ant define its directed distance to the endpoint it initially faces:

$$\delta_i = \begin{cases} x_i, & d_i = -1, \\ L - x_i, & d_i = +1. \end{cases} \quad (1)$$

In the pass-through system, ant i exits after time δ_i/v .

Theorem 1 (Exit-time multiset). *The unordered multiset of physical exit times under collision-and-reversal dynamics is*

$$\{\tau_1, \dots, \tau_n\} = \left\{ \frac{\delta_1}{v}, \dots, \frac{\delta_n}{v} \right\}.$$

Consequently, if $\delta_{(1)} \leq \delta_{(2)} \leq \dots \leq \delta_{(n)}$ are the ordered directed exit distances, then the k -th exit time is

$$T_{(k)} = \frac{\delta_{(k)}}{v}.$$

In particular, the completion time is

$$T = T_{(n)} = \frac{1}{v} \max_{1 \leq i \leq n} \delta_i. \quad (2)$$

Proof. By repeated application of the collision-invariance lemma, the interacting collision model has the same unordered set of occupied positions as the pass-through model at every time before exit. In the pass-through model, each trajectory moves in its initial direction until it reaches the corresponding endpoint, so its exit time is δ_i/v . Since collisions only exchange labels, the physical system has the same unordered multiset of exit times. Ordering those times gives $T_{(k)} = \delta_{(k)}/v$, and the last exit time is the maximum. $\square$

Maximum completion time

Since $0 \leq \delta_i \leq L$ for every i , equation (2) gives

$$T \leq \frac{L}{v}. \quad (3)$$

Thus

$$T_{\max} = \frac{L}{v} \quad (4)$$

whenever a configuration can place an ant arbitrarily close to one endpoint while facing the opposite endpoint. If endpoints are allowed and an ant beginning at an endpoint but facing inward is considered still on the stick, then the maximum is attained. If all initial positions are restricted to the open interval $(0, L)$, then L/v is the supremum rather than an attained maximum.

For the two-foot, one-foot-per-minute version,

$$T_{\max} = \frac{2}{1} = 2 \text{ minutes} = 120 \text{ seconds}.$$

Minimum completion time

The minimum depends on how the endpoints are treated. If ants may begin at endpoints and are counted as exiting immediately when placed at an endpoint, then

$$T_{\min} = 0.$$

If initial positions must lie strictly in $(0, L)$, then $T = 0$ is not attained, but the infimum is still 0.

Exercise 1 (Minimum completion time). Assume all ants begin at distinct points in the open interval $(0, L)$. Show that the completion time has no positive lower bound. In particular, for every $\varepsilon > 0$, construct a configuration of n ants with $T < \varepsilon$.

Solution. Choose distinct positions x_i satisfying $0 < x_i < v\varepsilon$ and direct every ant to the left. Then $\delta_i = x_i < v\varepsilon$ for all i , so

$$T = \max_i \frac{x_i}{v} < \varepsilon.$$

Therefore $\inf T = 0$, although no open-interval configuration attains $T = 0$.

5 Algorithms and Complexity

The direct physical simulation of the ant system requires tracking collisions and exits. The collision-invariance theorem eliminates this burden for the completion-time problem. The following algorithm computes the last exit time directly from the initial configuration.

Listing 1: Linear-time computation of completion time.

```
def completion_time(positions, directions, L, v):
    ans = 0.0
    for x, d in zip(positions, directions):
        if d == -1:
            delta = x
        else:
            delta = L - x
        ans = max(ans, delta / v)
    return ans
```

The runtime is $O(n)$ because the algorithm performs one constant-time computation per ant. This is also optimal in the standard input model; the argument follows the usual input-reading lower-bound logic used in asymptotic algorithm analysis.¹

Theorem 2 (Optimality for completion time). *Any deterministic algorithm that computes the completion time for arbitrary inputs must inspect every ant in the worst case. Therefore the completion time problem has a lower bound $\Omega(n)$, and the one-pass algorithm above is $\Theta(n)$.*

Proof. Suppose an algorithm does not inspect ant j . Construct two inputs that are identical on every inspected ant but differ in the uninspected ant j : in one input, ant j has a small directed exit distance; in the other, it has directed exit distance close to L . These two inputs can have different completion times while producing the same information to the algorithm. Thus any correct algorithm must inspect every ant in the worst case. The one-pass maximum computation matches this lower bound. $\square$

For the full ordered exit sequence, one must compute all directed exit times δ_i/v and sort them. This takes $O(n \log n)$ time by comparison sorting. In the comparison model, producing a fully ordered list of arbitrary real exit times requires $\Omega(n \log n)$ comparisons in the worst case, so the ordered exit sequence is computable in $\Theta(n \log n)$ time.¹

This contrast is the main algorithmic lesson: the full ordered process requires sorting, while the final completion time requires only a maximum.

6 Counting Collisions

Collisions do not affect the exit-time multiset, but they may still be numerous. Assume the initial positions are distinct and sorted so that

$$x_1 < x_2 < \dots < x_n.$$

In the pass-through picture, a physical collision occurs exactly when a right-moving trajectory from the left crosses a left-moving trajectory from the right. Therefore the collision count is

$$C = \#\{(i, j) : i < j, d_i = +1, d_j = -1\}. \quad (5)$$

This is an inversion statistic on the direction sequence.

Theorem 3 (Worst-case collision count). *The maximum possible number of physical collisions is*

$$C_{\max} = \left\lfloor \frac{n^2}{4} \right\rfloor.$$

Proof. If r ants on the left move right and ℓ ants on the right move left, then every right-left pair collides, giving $r\ell$ collisions with $r + \ell \leq n$. The product $r\ell$ is maximized when the two groups are as balanced as possible, so the maximum is $\lfloor n^2/4 \rfloor$. $\square$

If directions are independent fair coin flips, then for each ordered pair $i < j$ the event $(d_i, d_j) = (+1, -1)$ has probability $1/4$. By linearity of expectation,

$$\mathbb{E}[C] = \binom{n}{2} \frac{1}{4} = \frac{n(n-1)}{8}. \quad (6)$$

Thus the physical system can contain $\Theta(n^2)$ collisions even though the completion time is computable in $\Theta(n)$ time.

7 Random Initial Conditions

The deterministic theorem becomes more informative when positions and directions are random. Assume

$$X_i \sim \text{Uniform}(0, L), \quad \mathbb{P}(d_i = -1) = \mathbb{P}(d_i = +1) = \frac{1}{2},$$

independently across ants. The directed exit distance is

$$\delta_i = X_i \quad \text{if } d_i = -1, \quad \delta_i = L - X_i \quad \text{if } d_i = +1.$$

Since X_i and $L - X_i$ are both uniform on $[0, L]$, one has

$$\delta_i \sim \text{Uniform}(0, L).$$

Therefore the exit-time process is governed by order statistics of independent uniform random variables.^{7;3;2}

The k -th exit time

Let

$$0 \leq \delta_{(1)} \leq \delta_{(2)} \leq \dots \leq \delta_{(n)} \leq L$$

be the ordered directed exit distances. Then

$$\frac{\delta_{(k)}}{L} \sim \text{Beta}(k, n+1-k),$$

and hence

$$\frac{vT_{(k)}}{L} \sim \text{Beta}(k, n+1-k). \quad (7)$$

In particular,

$$\mathbb{E}[T_{(k)}] = \frac{k}{n+1} \frac{L}{v}.$$

The expected first exit time is therefore

$$\mathbb{E}[T_{(1)}] = \frac{1}{n+1} \frac{L}{v},$$

while the expected last exit time is

$$\mathbb{E}[T_{(n)}] = \frac{n}{n+1} \frac{L}{v}.$$

Completion-time distribution

Let $T = T_{(n)}$. For $0 \leq t \leq L/v$,

$$\mathbb{P}(T \leq t) = \mathbb{P}(\delta_1 \leq vt, \dots, \delta_n \leq vt) \quad (8)$$

$$= \left(\frac{vt}{L}\right)^n. \quad (9)$$

The density is therefore

$$f_T(t) = n \left(\frac{v}{L}\right)^n t^{n-1}, \quad 0 \leq t \leq \frac{L}{v}. \quad (10)$$

The moments are

$$\mathbb{E}[T^k] = \frac{n}{n+k} \left(\frac{L}{v}\right)^k, \quad (11)$$

so

$$\mathbb{E}[T] = \frac{n}{n+1} \frac{L}{v}, \quad (12)$$

and

$$\text{Var}(T) = \frac{n}{(n+1)^2(n+2)} \left(\frac{L}{v}\right)^2. \quad (13)$$

For the two-foot, one-foot-per-minute model,

$$\mathbb{E}[T] = \frac{2n}{n+1} \text{ minutes.}$$

For example,

$$n = 10 : \quad \mathbb{E}[T] = \frac{20}{11} \approx 1.818 \text{ minutes,}$$

while

$$n = 50 : \quad \mathbb{E}[T] = \frac{100}{51} \approx 1.961 \text{ minutes.}$$

As the number of ants grows, the expected completion time approaches the worst-case time of two minutes.

8 Asymptotics and Ants Remaining

Let

$$M_n = \frac{vT}{L}.$$

Under the uniform model, M_n is the maximum of n independent $\text{Uniform}(0, 1)$ random variables. Therefore

$$M_n \rightarrow 1$$

in probability as $n \rightarrow \infty$. More sharply, for $x \geq 0$,

$$\mathbb{P}(n(1 - M_n) > x) = \left(1 - \frac{x}{n}\right)^n \rightarrow e^{-x},$$

so

$$n \left(1 - \frac{vT}{L}\right) \Rightarrow \text{Exponential}(1). \quad (14)$$

Thus the gap between the completion time and the worst-case bound is typically of order $1/n$.

The same model also gives the number of ants remaining at time t . Define

$$R(t) = \#\{i : \delta_i > vt\}.$$

For $0 \leq t \leq L/v$,

$$R(t) \sim \text{Binomial}\left(n, 1 - \frac{vt}{L}\right). \quad (15)$$

Thus

$$\mathbb{E}[R(t)] = n \left(1 - \frac{vt}{L}\right), \quad (16)$$

and

$$\text{Var}(R(t)) = n \left(1 - \frac{vt}{L}\right) \left(\frac{vt}{L}\right). \quad (17)$$

This gives a direct theoretical explanation for an interactive simulation counter showing how many ants remain on the stick at each time.

9 General Initial Distributions

The uniform model is only one special case. Suppose the initial positions are iid with continuous cumulative distribution function F on $[0, L]$, and suppose directions are independent with

$$\mathbb{P}(d_i = -1) = p, \quad \mathbb{P}(d_i = +1) = 1 - p.$$

Then the directed exit distance δ_i has CDF

$$G(y) = pF(y) + (1 - p)[1 - F(L - y)], \quad 0 \leq y \leq L. \quad (18)$$

Indeed, the first term corresponds to left-moving ants with $X_i \leq y$, while the second corresponds to right-moving ants with $L - X_i \leq y$, or equivalently $X_i \geq L - y$.

Theorem 4 (General completion-time distribution). *Under the assumptions above, the completion time satisfies*

$$\mathbb{P}(T \leq t) = G(vt)^n, \quad 0 \leq t \leq \frac{L}{v}, \quad (19)$$

where G is defined in equation (18).

Proof. By the exit-time theorem, $T \leq t$ if and only if every directed exit distance satisfies $\delta_i \leq vt$. Since the δ_i are iid with CDF G ,

$$\mathbb{P}(T \leq t) = \prod_{i=1}^n \mathbb{P}(\delta_i \leq vt) = G(vt)^n.$$

□

For $F(y) = y/L$ and $p = 1/2$, one obtains $G(y) = y/L$, recovering the uniform model. This general theorem is useful because it separates the collision dynamics from the input distribution. Once the directed-distance CDF is known, the completion-time distribution follows immediately.

10 Simulation and Verification

The theory suggests two computational approaches. The first simulates the physical system directly by finding collision and exit events. The second ignores collisions and computes the directed exit times. The second method is not an approximation; it is an exact reduction.

Listing 2: Reduced JavaScript computation for completion time and exit sequence.

```
function directedExitTimes(positions, directions, L, v) {
  return positions.map((x, i) => {
    const delta = directions[i] === -1 ? x : L - x;
    return delta / v;
  });
}

function completionTime(positions, directions, L, v) {
  const times = directedExitTimes(positions, directions, L, v);
  return Math.max(...times);
}

function orderedExitTimes(positions, directions, L, v) {
  return directedExitTimes(positions, directions, L, v)
    .sort((a, b) => a - b);
}
```

A direct simulator is still useful for visualization. It shows the apparent complexity of collisions, while the reduced computation provides a correctness oracle. A direct collision simulator and the reduced pass-through computation should agree on the unordered exit-time multiset and the final completion time. If they do not, then the simulator likely contains an error in collision handling, event ordering, or endpoint processing.

For Monte Carlo validation under the uniform model, one may generate many independent configurations, compute T , and compare the empirical CDF with

$$F_T(t) = \left(\frac{vt}{L}\right)^n.$$

Such simulation is not needed to prove the result, but it is a useful computational check and a natural way to connect the theorem to an interactive demonstration.^{6;7}

11 Risk-Style Interpretation and Applications

The ant system is not a literal financial-market model. It contains no order book, price process, liquidity constraint, strategic behavior, or stochastic volatility. Still, it is a useful stylized model of a time-to-clear system: completion occurs only when the slowest directed component exits. The relevant variable is therefore an extreme value,

$$T = \max_i \frac{\delta_i}{v}.$$

This makes the problem a compact example of bottleneck risk.

Under the uniform model, the q -quantile of completion time is

$$Q_q(T) = \frac{L}{v} q^{1/n}. \quad (20)$$

A VaR-style time-to-clear measure may therefore be written as

$$\text{VaR}_q(T) = \frac{L}{v} q^{1/n}. \quad (21)$$

The corresponding expected tail completion time is

$$\mathbb{E}[T \mid T > Q_q(T)] = \frac{L}{v} \cdot \frac{n}{n+1} \cdot \frac{1 - q^{(n+1)/n}}{1 - q}. \quad (22)$$

This is analogous in spirit to expected shortfall: it averages completion times conditional on being in the upper tail.⁸

The defensible applications are methodological rather than literal. The analysis illustrates:

- **Algorithmic simplification:** a collision-heavy physical process reduces to a one-pass maximum computation.
- **Simulation verification:** the reduced formula provides an exact oracle for testing event-driven collision simulators.
- **Bottleneck modeling:** system completion is governed by the slowest directed exit distance.
- **Time-to-clear risk:** exact quantiles and tail expectations can be derived for a stylized completion-time variable.
- **Interactive education:** the problem gives a visual example of how invariance can replace brute-force simulation.

The limitations are equally important. The reduction requires equal speeds, point particles, instantaneous collisions, and unlabeled observables. It does not model named-agent trajectories, finite body lengths, strategic behavior, queueing, or real financial market microstructure.

12 Conclusion

The ants-on-a-stick puzzle appears to be a many-body collision problem, but its exit-time structure is governed by a simple invariant. Collision-and-reversal is equivalent, for unlabeled exit times, to pass-through motion with relabeling. Consequently, the full exit-time multiset is exactly

$$\left\{ \frac{\delta_1}{v}, \dots, \frac{\delta_n}{v} \right\},$$

and the completion time is

$$T = \frac{1}{v} \max_i \delta_i.$$

The worst-case completion time is L/v , while the open-interval minimum is not attained but has infimum zero.

The algorithmic consequences are sharp: completion time is computable in optimal $\Theta(n)$ time, even though the physical collision count can be $\Theta(n^2)$. When positions and directions are random, the exit process becomes an order-statistic problem. This yields exact distributions, expectations, variances, quantiles, asymptotic laws, and remaining-agent counts. The result is therefore more than a puzzle trick. It is a compact example of how an interaction-heavy system can collapse into a deterministic metric, an optimal algorithm, and an exact stochastic completion-time model.

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